

Effects of livestock farming on birds of rural areas in Europe

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Received: 21 July 2015 / Revised: 15 March 2016 / Accepted: 17 March 2016 /

Published online: 25 March 2016

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Abstract In the last decades, profound modifications of agricultural practices occurred in Europe, including the introduction of modern livestock farming. These modifications negatively affected the fauna of rural areas, as indicated by the large demographic declines suffered by several populations of birds typical of these habitats. The impact of agricultural practices on bird populations has been widely investigated, while the effect of livestock farming has seldom been assessed. To fill this gap, we carried out a quantitative meta-analysis of the existing scientific literature and evaluated the size of the effects of livestock farming on birds of rural areas in Europe. We only found 26 papers on this topic, from which 72 effect sizes could be estimated. The barn swallow (*Hirundo rustica*) was the species on which most studies focused. Livestock farming positively influenced presence and distribution of barn swallows in breeding habitats, while it did not significantly affect reproduction of this species. Effects on other bird species typical of rural habitats were non-significant. The positive effect on the insectivorous barn swallow might be mediated by the enhanced insect abundance where livestock is reared. In addition, habitat features typical of rural settings where livestock is reared (e.g. cattle-sheds or large hayfields) positively affected barn swallows independently of actual presence of livestock at a setting. Presence of livestock at rural setting therefore seems beneficial to barn swallows, but not

Communicated by Karen E. Hodges.

Electronic supplementary material The online version of this article (doi:[10.1007/s10531-016-1087-9](https://doi.org/10.1007/s10531-016-1087-9)) contains supplementary material, which is available to authorized users.

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significantly to other bird species typical of rural habitats. The effect of livestock farming on birds of rural habitats has been under-investigated to date.

Keywords Agricultural intensification · Barn swallow · Farmlands · Hierarchical linear models · Livestock farming · Meta-analysis

Introduction

In the last centuries, increasing human populations caused large-scale clearance of woodlands and the expansion of land used for agriculture worldwide (Meyer and Turner II 1992). In Europe, extended agricultural habitats favoured several bird species that use rural habitats as breeding, wintering or staging sites. Some species have become so typical of farmland that, irrespective of their phylogenetic relationships, they are now grouped under the term “farmland birds” (Harrison 1988; Lefranc 1997). Despite being an artificial habitat, European farmlands support nearly 120 bird species of conservation concern, most of which are exclusively dependent on this habitat for breeding or wintering (Tucker 1997).

The changes that occurred in farming practices since the middle of the last century probably negatively affected the fauna of rural habitats (Anderson et al. 2001; Donald et al. 2002; Fuller et al. 1995; Gregory et al. 2005). For example, European farmland birds showed stronger demographic declines than birds of other habitats during the last decades (Fuller et al. 1995; Donald et al. 2001, 2006), and overall their populations were reduced by 57 % between 1980 and 2013 (PECBMS 2015). Such changes include, among others, the mechanization of agriculture and the extension of arable fields at the expense of less productive areas, such as hedgerows, field boundaries and pastures, which, in contrast, are suitable habitats for foraging and nesting by birds (Lefranc 1997; Gillings and Fuller 1998; Evans et al. 2003; Henderson et al. 2007), the simplification of crop rotation, which caused a reduction of mixed farming systems and an increase in cereal and silage production (Vickery et al. 2001), an increase in the use of fertilizers and pesticides (Geiger et al. 2010) and changes in grassland management (Donald et al. 2002). The resulting simplification of farmland habitats, in turn, probably caused a reduction in food availability for birds, particularly invertebrates and seeds, and of nesting sites, with deleterious effects on their populations (Benton et al. 2003; Eggers et al. 2011).

Several studies have investigated the effects of changes in field management on birds (Evans and Robinson 2004; Evans et al. 2007; Henderson et al. 2007). However, these changes often occurred in parallel with changes in animal husbandry since farms often tended to specialize in either arable crops or livestock (Robinson and Sutherland 2002). In particular, rearing practices shifted from traditional ones, where a small number of heads is reared in a relatively large number of farms, to modern ones, where hundreds or even thousands of heads are reared in a relatively small number of large, industrialized and factory-like zootechnical plants. Such changes in zootechnical practices, in particular the abandonment of grazing, may also have contributed to a decrease in the suitability of farmland ecosystems for birds and, consequently, may have caused their decline (Evans et al. 2005).

In this study, we aimed at quantifying the magnitude of the effects of livestock farming practices on the distribution and reproduction of birds of rural habitats in Europe. To this end, we systematically reviewed the scientific literature on this topic and performed a

quantitative meta-analysis. In particular, we considered studies investigating the effects of both indoor livestock husbandry, such as cow-rearing in cattle-sheds, and outdoor ones, such as extensive grazing on pastures.

While reviewing the literature, we observed that the majority of the studies on the effects of livestock farming on birds of rural habitats focused on a single species: the barn swallow (*Hirundo rustica*). Indeed, several studies related the abundance, distribution and breeding success of barn swallows to the presence of livestock farming, which is considered an important ecological factor that influences both food availability and quality of nest sites for this species (Møller 1983, 2001; Ambrosini et al. 2002a, b; Ambrosini and Saino 2010; Evans and Robinson 2004; Turner 2006; Gruebler et al. 2010; Orłowski and Karg 2011, 2013; Willi et al. 2011). Consequently, in this meta-analysis, besides investigating the effect of livestock farming on birds of rural habitats (see “Methods” section), we focused particularly on the barn swallow, because the largest amount of information is available on this species.

Presence of livestock may have both direct and indirect effects on birds. We considered as direct effects the changes in the distribution or reproductive success of birds observed when livestock was reared or when there was grazing at a site when the study was conducted. These effects are thought to be mainly due to a larger abundance of the invertebrate fauna at, or very close to, the study site due to livestock presence. Indeed, the invertebrate fauna is important not only for strictly insectivorous species, but also for other species during chick rearing (Newton 2004; Henderson et al. 2007; McMahon et al. 2010; Orłowski and Karg 2011, 2013; Willi et al. 2011). In contrast, we considered as indirect effects the changes in the distribution or reproductive success of birds that were observed in a study, but linked to environmental or habitat features of settings where livestock was reared, but that do not imply presence of animals at the setting when the study was conducted. Such indirect effects include, for instance, larger availability of favourable nesting sites for barn swallows and other species (e.g. European starling *Sturnus vulgaris*) in a farm due to the presence of rural buildings devoted to animal rearing, such as traditional cattle-sheds, piggeries and stables, which often remain accessible to birds even when livestock is no longer reared at that farm (Møller 2001; Turner 2006; Gruebler et al. 2010; Ambrosini and Saino 2010; van den Brink 2003; Willi et al. 2011; Von Post and Smith 2015). Similarly, hayfields and meadows, which are the preferred foraging habitats of barn swallows and other birds, are usually larger around farms where livestock is reared compared to farms with no livestock (Møller 2001; Ambrosini et al. 2002a; Evans et al. 2007; Gruebler et al. 2010; Sicurella et al. 2014). Importantly, differently from pastures, hayfields and meadows are not regularly grazed, but are left growing for hay production. Since livestock is usually absent from these fields, we consider their effects on birds as indirect ones. Such indirect effects are expected to contribute to improving the overall habitat suitability for birds. In this study we therefore evaluated separately direct and indirect effects of livestock presence on birds. Further details on the distinction between direct and indirect effect is provided in the Methods section.

Effects of presence of livestock farming in agricultural settings may affect both distribution and reproduction of birds. In this meta-analysis, we therefore assessed and compared the size of the effects of livestock on bird distribution by cumulating studies considering presence and abundance of bird species (‘distribution’ studies hereafter, see also Methods). Similarly, we evaluated the effect size of livestock farming on different estimates of the reproductive success of birds (e.g. clutch or brood size, fledging success, ‘reproduction’ studies hereafter, see also “Methods” section).

Overall, we expect that the presence of livestock is beneficial for birds (i.e. determines larger abundance and larger breeding success). In addition, we expect that both direct and indirect effects of livestock farming are beneficial for birds, with direct effects having a larger effect size than indirect ones.

Methods

Literature search

On October 31st 2015, we searched for published scientific papers that investigated the effects of livestock farming and grazing on birds of rural habitats in Europe. Our analyses focused on the 39 bird species included in the list of European common farmland birds (PECBMS 2015) and on the house martin, *Delichon urbicum*, the only species not listed among farmland birds but that commonly occurs in European farmland landscapes and for which we found one study relating its distribution to livestock farming (Willi et al. 2011). Hereafter we will refer to the group of species included in this meta-analysis as “birds of rural habitats”. We used the search platforms *Web of Science* and *Google Scholar* with the following keywords: “bird”, “farmland bird”, “farming”, “rearing”, “livestock”, “agricultural intensification”, “grazing” and “pasture”. Moreover, we searched for papers that focused on farmland birds and on the house martin using their Latin scientific and common English name (e.g. “*Hirundo rustica*” and “barn swallow”). We also used a combination of keywords with “bird” or the scientific names of species plus another keyword among the previous ones (e.g. “*Delichon urbicum*” + “livestock”). Although “agricultural intensification” was among the keywords, we carefully evaluated the papers containing this keyword in order to select only those reporting information about the presence of livestock or about presence of environmental features typical of livestock farming settings (e.g. traditional cattle-sheds; see Introduction and below for details). Finally, we checked the reference list of the resulting articles and of all the papers citing each publication resulting from this search in order to identify other papers that could be relevant for our review.

We limited the review to papers from Europe. In detail, we considered studies from Denmark, Germany, Hungary, Norway and Russia (1 paper each), Switzerland and Poland (2 papers each), the Netherlands (3 papers), the United Kingdom (6 papers) and Italy (8 papers). Studies from outside Europe were excluded because we aimed at including only studies conducted on a homogeneous group of bird species (European farmland birds plus the house martin). In addition, only five papers from other continents addressed the topic of the effect of livestock farming on birds (details not shown). We also excluded papers for which we could not assess size of the effects on individual species of birds of rural habitats. For example, we excluded the paper by Pavel (2004) because it compared nest failure due to trampling by grazing animals by cumulating data on two farmland species (the meadow pipit *Anthus pratensis* and the skylark *Alauda arvensis*) and on a non-farmland bird (the water pipit *Anthus spinoletta*).

Livestock species considered in the studies included in the meta-analysis were mainly cows, horses, pigs and sheep. Only two studies (van den Brink 2003, Orłowski and Karg 2013, both included in the meta-analysis) investigated the effects of poultry farming on birds, while some studies did not report separate effects of each livestock species, but only that of presence of livestock (e.g. they reported the effect of presence of either cattle, pigs

or horses on a farm, as in Sicurella et al. (2014)). For this reason, we did not attempt to investigate separately the effects of different species of livestock on birds.

Since 15 out of the 26 reviewed papers that reported the effects of livestock farming on bird species focused on the barn swallow (see “Results” section and Table S1), we investigated separately the effects of livestock farming on this species and on the other birds of rural habitats.

Effect size estimates

We extracted statistics on effects of livestock farming from all the results reported in the selected publications and transformed them into Pearson’s product-moment correlation coefficients (Pearson’s r) that was used as a measure of effect size because it indicates the strength of the relationship between two variables. More precisely, its squared value r^2 provides information about the proportion of variance explained by the predictor variable in a sample (Rosenthal 1991, 1994). In accordance with Cohen’s rule, a ‘small’ effect size corresponds to a Pearson’s r of 0.10 (i.e. $r^2 = 1\%$ of the variance), a ‘medium’ effect size corresponds to an r of 0.30 (i.e. $r^2 = 9\%$ of the variance) and a ‘large’ effect size corresponds to an r of 0.5 (i.e. $r^2 = 25\%$ of the variance) (Cohen 1988). Since r -values do not follow a normal distribution, they were Z-transformed (Z_r) before the analyses using Fisher’s transformation (Fisher 1928).

Where available, we extracted from the paper the absolute values of all Z_r or r coefficients and the relative sample sizes. Hence, a ‘paper’ may report one or more effect sizes. The direction (sign) of the effect was then assigned based on the congruence of the results with the hypothesis that birds of rural habitats benefit from the presence of livestock farming. If the result of an analysis supported the hypothesis (e.g. birds were more abundant or had larger reproductive success where livestock was reared), we assigned a positive sign to the effect size. Conversely, if the result did not support this hypothesis, we assigned a negative sign. When studies reported statistics different from Z_r or r coefficients, they were included as follows. The test statistics F , χ^2 and t were first transformed into r -values using the equations proposed by Rosenthal (1991) and then into Z_r using Fisher’s transformation (e.g. $F_{1,91} = 5.9$ from Ambrosini et al. 2002a was transformed into $r = -0.2468$ and $Z_r = -0.2520$; this effect size was given a negative sign because reproductive success was lower in farms with than without livestock). When no statistics were available or non-parametric tests were reported, we computed effect sizes from P values (e.g. $P = 0.0457$, sample size = 24 from Evans et al. 2015 was first transformed into a one-tailed p value, i.e. $P = 0.023$, and then into $r = 0.4031$ and $Z_r = 0.4273$). Alternatively, we used information about means and standard deviations reported in the studies to estimate effect sizes (e.g. mean number of nests in organic farms 2.40 ± 3.38 SD, sample size = 20 and in conventional farms 2.55 ± 4.50 SD, sample size = 20 from Kragten et al. (2009) were transformed into $r = -0.0263$ and $Z_r = -0.0263$, with a negative sign because of the larger number of nests in conventional than in organic farms). When only approximations of P -values were reported, we considered the extreme upper bound of the range (i.e. for $P < 0.001$, we used $P = 0.001$ and transformed this value into r and Z_r as above) to compute the effect size. Hence, we were conservative in the estimation of effect sizes in these cases.

When the sign of the effect was unclear, we first tried to extract this information from other parts of the paper (e.g. discussion or graphs). If this was not possible, we asked the authors for further information. Similarly, we asked the authors for information on the size of the effects, when a paper reported undefined values (e.g. ‘ns’ or ‘ $P > 0.05$ ’). If the

authors did not reply to our first request of information, we sent them a second request. If we received no answer to this second request, we excluded the effect size from the meta-analysis. Overall, only two effect sizes from two different papers were excluded for this reason (details not shown).

All the effect sizes that we could extract from each paper were included in the meta-analysis because we used a statistical approach able to account for multiple effect sizes from the same paper (see below). The sample units of the results from which we calculated effect sizes included in the meta-analyses were alternatively breeding pairs, colonies, buildings, farms, fields or even years (see Table S1 for details on sample unit at each study). However, due to the small number of studies found in the literature on the effect of livestock farming on birds, we cumulated all effects sizes irrespective of the sample unit.

Direct and indirect effects and effects on bird distribution and reproduction

We aimed at evaluating separately direct and indirect effects of livestock farming on birds (see “[Introduction](#)” section). We therefore ran separate meta-analyses on the effect sizes from analyses investigating the influence of: (1) presence of reared animals per se (direct effects); and (2) presence of environmental features typical of livestock farming settings (indirect effects). In detail, effect sizes from analyses relating bird distribution or reproduction to presence or abundance of livestock, presence of livestock farming activities, grazing intensity and insect abundance were considered as deriving from direct effects of livestock on birds. In contrast, effect sizes from analyses reporting the effects on birds of presence of buildings devoted to animal rearing, presence of hayfields, meadows or other non-grazed grasslands used to produce hay, or other features of the farmland habitats linked to the presence of livestock (e.g. presence of manure heaps), including the presence of organic farming practices (Lubbe and de Snoo 2007; Kragten et al. 2009), were considered as deriving from indirect effects of livestock on birds. Indeed, organic farming usually implies the use of manure as the only fertilizer (Kragten et al. 2009).

When a paper included analyses about different livestock species (e.g. Orłowski and Karg 2011), we included one effect size for each livestock species, and accounted for pseudo-replication of effect sizes from the same paper in the analyses (see below). If a paper reported only one effect size about the overall effect of animal husbandry, without any distinction among livestock species (e.g. Hart et al. 2002), we reported that single value.

In this study, we were interested in investigating the influence of livestock farming on the presence and on the reproduction of birds in rural habitats. However, papers sometimes reported different analyses dealing with ‘distribution’ or ‘reproduction’ of a species. For example, Ambrosini et al. (2002a) reported effects of livestock farming on both presence and size of barn swallow colonies. In these cases, we included in the meta-analysis only the effect sizes calculated from the analyses that provided more information on either distribution or reproductive success of birds. For instance, when both analyses on presence and abundance of a species were reported in the same paper, we only included effect sizes from analyses on species abundance and discarded those from analyses on presence. Similarly, effect sizes from analyses investigating fledgling success were preferred over those from analyses investigating brood size, which in turn were preferred over those from analyses on clutch size.

After this selection, we obtained 72 effect sizes from 26 papers. Table S1 reports details on all effects sizes (variance, sample size and unit, direction of the effect) and additional

information (bird species, country, latitude and habitat type of the study area) as well as full reference and year of publication of each paper reporting effect sizes.

Meta-analyses

Different effect sizes extracted from the same paper cannot be considered statistically independent. We therefore used hierarchical linear models (aka linear mixed effect models, LMMs) assuming a Gaussian error distribution. Indeed, LMMs are able to account for non-independence in the data and their use for meta-analyses is advocated precisely for analyses including multiple effect sizes from the same paper (Van den Noortgate et al. 2013). In this case, each effect size (Z_r) was weighted by its variance (equal to sample size -3), while the random part of the model included paper ID and effect size ID, nested within paper ID, as random grouping factors. In addition, residual variance was constrained to be equal to 1 in all models. With this parameterization, we obtained a mixed model able to fit a three-level meta-analysis (*sensu* Van den Noortgate et al. 2013). The reasons for such a parameterization are fully described in Van den Noortgate et al. (2013).

We first investigated whether mean size of the effects of livestock farming on the barn swallow, and on the other bird species differed significantly from zero by running separate LMMs including only the intercept as fixed predictor (Van den Noortgate et al. 2013). In these models, sizes of direct and indirect effects, and of effects on distribution and reproduction were considered together. Species was entered as a further random grouping factor (crossed with effect size ID and paper ID) in all the models run on bird species others than the barn swallow. Second, we investigated in separate meta-analyses whether mean size of direct and indirect effects differed from zero for both the barn swallow and the other bird species. In these analyses, we considered both effects on distribution and reproduction together, hence we ran four additional LMMs with the same fixed and random structure. Third, we evaluated separately whether mean size of the effects of livestock on bird distribution and on barn swallow reproduction differed from zero. We did not run any analysis on the effects of livestock farming on the reproduction of other bird species because we only found three effect sizes from studies investigating this topic. Therefore, we ran three additional models with the same fixed and random structure as above. Fourth, we compared the size of direct and indirect effects and of effects on barn swallow distribution and reproduction by running LMMs while including as predictors dichotomous moderator variables indicating whether a study investigated direct or indirect effects, or effects on the barn swallow or on other species.

Analyses were run with the SAS 9.4 proc MIXED (SAS Institute Inc, Cary, NC) by a restricted maximum likelihood (REML) procedure. Residual variance was set to 1 by the PARMs statement as is necessary in meta-analyses run on Z_r values. Degrees of freedom were estimated with the Satterthwaite approximation. Since the separate meta-analyses revealed heterogeneity of variances between levels of the moderator (e.g. between direct and indirect effects), we allowed for heterogeneity of variances at the study and at the paper levels by the GROUP option in the RANDOM statement. We note that with these settings, marginal means estimates from the LMMs with moderator variables obtained by the LSMEANS statement gave identical results as mean estimates from the separate meta-analyses. However, these allowed comparing mean size of the effects between levels of the moderator variable.

In the Supporting Information, we provide the code for all the analyses.

Influence of latitude and year of publication

We investigated whether bird populations living at different latitudes were differently affected by livestock practices, by relating effect sizes to the latitude of the study areas. When the exact coordinates were reported, we rounded them to the nearest degree. If no coordinates were reported in the paper, we searched for the study area on Google Earth™ (Google Inc., Mountain View, CA) and considered the latitude (rounded to the closest degree) of its approximate centre. If a study was conducted in several locations or across a wide area, we calculated the mean value between the northernmost and southernmost latitudes of the study area(s) and rounded this value to the closest degree.

Moreover, we evaluated whether the effect sizes changed in accordance with year of publication of the paper, as expected if studies with large effects were likely to be published first, while studies with small effects were likely to take longer time to be published (Møller and Jennions 2001). For these purposes we ran two distinct LMMs where latitude and year of publication were the predictor variables, and the effect size ID and paper ID (accounting for possible pseudo-replications) were entered as random factors. These analyses were also replicated by considering either the effect sizes on the barn swallow or those on the other species. In these latter analyses, species was entered as a further random grouping factor.

Publication bias analyses

Studies reporting null or non-significant results may remain unpublished and therefore do not appear in the scientific literature. Such a deficit, if existing, may lead to an overestimation of significant studies and can bias the average effect size resulting from a meta-analysis (Song et al. 2000; Møller and Jennions 2001). To partly account for this potential issue, we examined indirect evidence for the presence of publication bias in our dataset using different statistical methods. First, we used the Kendall rank correlation test between effect sizes and sample sizes across all the 72 effect sizes (Begg and Mazumdar 1994). A high correlation value would indicate that the funnel plot of effect sizes on sample sizes is asymmetric, which may be due to publication bias. Second, we estimated the fail-safe number using Rosenthal's method with alpha equal to 0.5 (Rosenthal 1979). This test estimates the number of studies with an effect size equal to zero needed to make the mean effect size not significantly different from zero. Third, we estimated the number of possible 'missing' studies using the 'trim and fill' method with L_0^+ estimator (Duval and Tweedie 2000). In this analysis, we assumed that the funnel plot of effect sizes on sample sizes should be symmetric to the mean effect size in absence of publication bias. The model determines the number of 'missing' studies that would produce a symmetric funnel plot and finally estimates the new mean effect size that would be obtained by adding these studies to the original sample. These analyses were run by the *metafor* library (Viechtbauer 2010) in R 3.0.3 (R Core Team 2014).

We ran the publication bias analyses only when the mean effect sizes resulting from the models differed significantly from zero (see Table 1) because we aimed at assessing whether publication bias may have introduced Type-I errors in our analyses.

Our meta-analyses included different effect sizes obtained from the same paper, which are therefore non-independent. The LMMs we used for the analysis accounted for this non-independence of effect sizes (see 'Meta-analyses' above), but we are aware of no procedure for the test of publication bias that accounts for non-independence of effect sizes. We

Table 1 Results from summary meta-analyses

Effect	N	Z_r	SE	t	df	95 % CI	P
Barn swallow							
All	47	0.272	0.051	5.38	43.3	0.170–0.374	<0.001
Direct	33	0.337	0.078	4.35	7.89	0.158–0.516	0.002
Indirect	14	0.147	0.039	3.82	5.23	0.049–0.245	0.011
Distribution	39	0.297	0.055	5.41	35.6	0.185–0.408	<0.001
Reproduction	8	0.156	0.176	0.89	2.9	−0.163–0.200	0.418
Other bird species of rural habitats							
All	25	0.043	0.129	0.33	7.14	−0.260–0.346	0.749
Direct	17	0.054	0.138	0.39	7.37	−0.268–0.377	0.705
Indirect	8	0.081	0.158	0.51	2.98	−0.424–0.586	0.645
Distribution	22	0.080	0.129	0.62	5.94	−0.237–0.397	0.558
Reproduction	3	–	–	–	–	–	–

Number of effect sizes (N), mean effect size (Z_r) weighted by their variance (equal to sample size -3), standard error (SE), t -value with the relative degrees of freedom (df), 95 % confidence interval, and P -values are reported. Analyses were performed by considering all studies (all) and separately for direct or indirect effects of livestock farming and for studies dealing with distribution or reproduction of the barn swallow or of the other species of rural habitats

therefore repeated the publication bias analyses twice. First, we ran the analyses by considering all effect sizes as independent. Second, we calculated the mean Z_r value within each paper by running random effect meta-analytic models on all the Z_r values from the same paper with the *metafor* package in R, and then ran the publication bias analyses by including the mean Z_r value for each paper reporting more than one effect size. This latter procedure corresponds to considering all effect sizes reported in the same paper as repeated measures of the same effect size. When both the analysis run considering multiple effects sizes from the same study as fully independent and that run considering them as fully dependent were non-significant, we concluded that publication biases did not affect our conclusions.

When running the publication bias analyses on mean within-paper Z_r values, we were faced with the problem of assessing the correct sample sizes. For example, the same paper may report multiple effect sizes calculated on the same sample of farms or on different samples of farms. Even more complex situation may occur. For example, breeding success can only be estimated for the subsample of farms where barn swallows were present among a random sample of farms in an area, which can both host breeding swallows or not (e.g. Ambrosini et al. 2002a). Importantly, sometimes papers were not crystal-clear in reporting which analysis was run on which subset of sample units. This sometimes determined the impossibility of properly assessing an overall sample size to be attributed to each paper (but note that we could always identify the sample size for all the analyses from which we calculated individual effect sizes). To solve this problem, we calculated the mean, minimum and maximum sample size among all the Z_r values we mediated to run publication bias analyses on mean within-paper effect sizes. We then re-ran the Kendall's tau correlation test by using each of these within-paper estimates of effect sizes, and compared the results. We note that the analyses of the fail-safe number and the trim-and-fill number of missing studies can be run by including in the model the variance of the mean within-paper

effect size obtained by the random effect model used to calculate it, so that these analyses do not require estimating within-paper sample sizes.

In the Supporting Information, we report the R code for running all the publication bias analyses.

Results

Reviewed studies

We identified 26 published papers reporting statistics suitable for a meta-analysis. Fifteen (58 %) papers focused on the barn swallow only, ten (38 %) on other species but the barn swallow, one paper (4 %) focused on both the barn swallow and the house martin. From these papers, we extracted 72 effect sizes (reported in Table S1). Forty-seven of them (65 %) were about the barn swallow, 33 (70 %) and 14 (30 %) of which reported respectively direct and indirect effects. In addition, 39 (83 %) and eight (17 %) of these effect sizes focused respectively on distribution and on reproduction of the barn swallow. Twenty-five effect sizes (35 %) were about other birds of rural habitats, 17 (68 %) reported direct effects, eight (32 %) indirect effects, 22 (88 %) effects on distribution and three (12 %) effects on reproduction. In detail, studies included in the present meta-analysis were on the house martin and on eight of the 39 species classified as European farmland birds (i.e. meadow pipit *Anthus pratensis*, northern lapwing *Vanellus vanellus*, red-backed shrike *Lanius collurio*, skylark *Alauda arvensis*, starling *Sturnus vulgaris*, whinchat *Saxicola rubetra*, white stork *Ciconia ciconia*, and yellow wagtail *Motacilla flava*; see Table S1).

Effect of livestock farming on farmland birds

Livestock farming had an overall medium effect ($r = 0.265$) on barn swallows accounting on average for 7.0 % of the observed variance (Table 1 and Fig. 1). The mean size of direct effects was also medium ($r = 0.325$), explaining 10.5 % of the variance, while that of indirect effects was small ($r = 0.147$), explaining only 2.1 % of variance (Table 1). However, all these effects were significantly larger than zero (Table 1), and a meta-analysis with moderator variable revealed that mean size of direct effects of livestock farming on the barn swallow did not differ significantly from that of indirect ones (difference: $Z_r = 0.190 \pm 0.087$ SE, $t_{11,2} = 2.19$, $P = 0.051$), albeit the difference was only marginally non-significant. This suggests that presence of livestock (direct effects) tended to have, on average, slightly larger effects on barn swallows than environmental or habitat features of settings linked to livestock presence, but that do not imply current presence of animals (indirect effects, see Introduction), and that both direct and indirect effects of livestock are beneficial to this species.

Livestock farming (both direct and indirect effects) had a positive and medium-sized effect on barn swallow distribution ($r = 0.289$) explaining 8.3 % of the variance, but did not significantly affect barn swallow reproduction ($r = 0.156$ corresponding to 2.4 % of variance explained; Table 1). The size of the effects on barn swallow distribution did not differ significantly from that of effects on reproduction of this species (difference: $Z_r = 0.140 \pm 0.184$ SE, $t_{3,49} = 0.76$, $P = 0.494$).

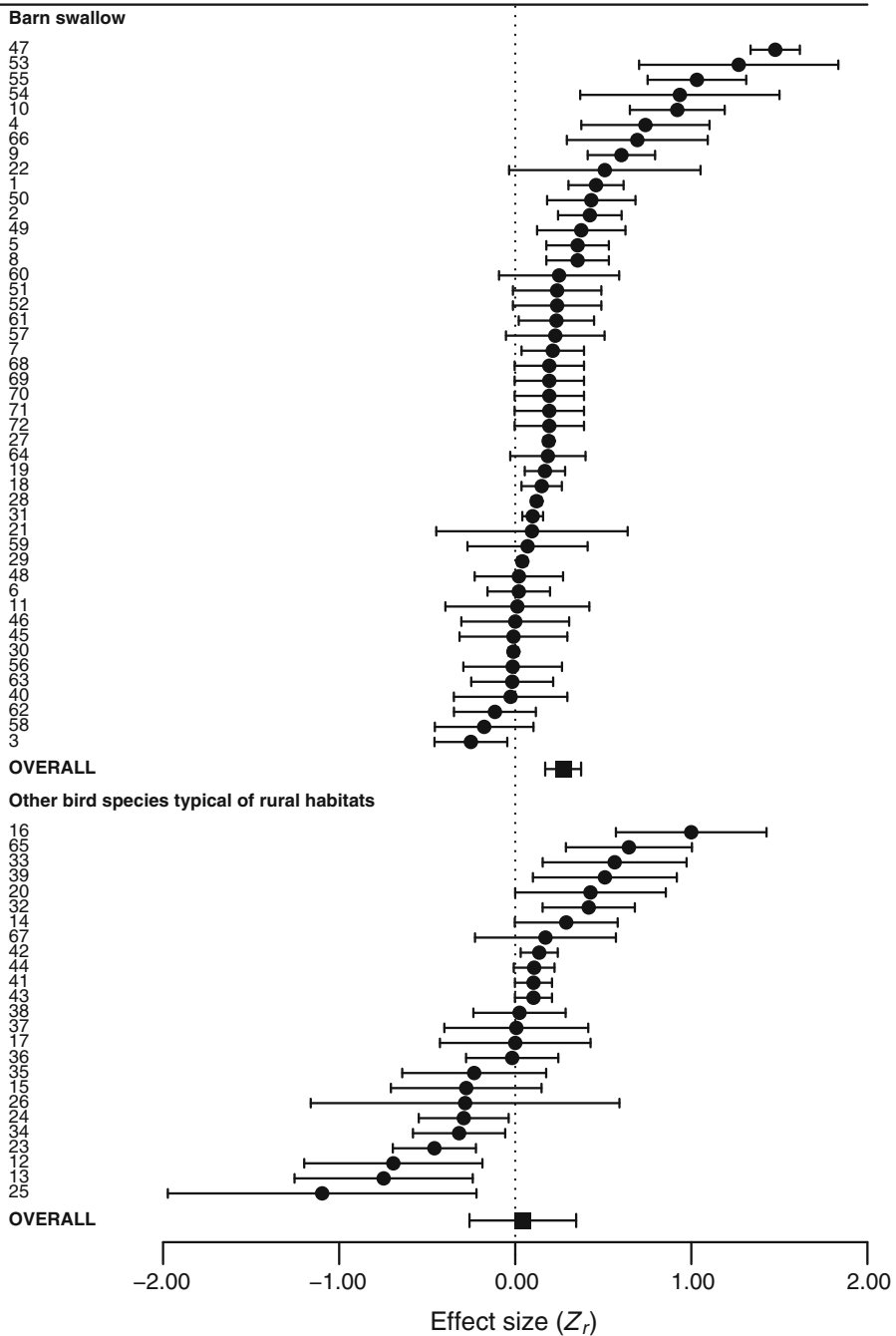


Fig. 1 Forest plots of effect sizes. Forest plots (dots) and their 95 % confidence intervals are reported for the 72 studies included in the meta-analysis. Studies on the barn swallow and on the other species of rural habitats are reported separately. Numbers on the left refer to effect size ID as reported in Table S1. The overall effects for each group of studies (squares) calculated from the meta-analysis (lines with “All” in Table 1) are reported with their 95 % confidence intervals

When other bird species of rural habitats than the barn swallow were considered, no significant effect of livestock farming was observed for any measure (Table 1 and Figure S1). The paucity of studies on reproduction of these species (equal to three) did not allow any analysis.

Effect size did not change according to either latitude ($|I_{8.02}| \leq 0.96$, $P \geq 0.363$) or year of publication ($|I_{7.89}| \leq 1.20$, $P \geq 0.266$), and this held true when we considered either studies on the barn swallows or on all the other species but the barn swallow.

Tests of publication bias

The results from publication bias analyses were similar when run on all effect sizes separately (i.e. assuming independence of effect sizes reported in the same paper) and on mean effect sizes within a paper (i.e. assuming that effect sizes reported in the same paper are replicated measures of the same effect size) (Table S2).

In the analyses of the barn swallow, the Kendall correlation test was significant and negative only when run on mean within-paper effect-sizes on reproduction with sample size set equal to the minimum sample size within papers (Kendall's $\tau = -0.516$, $P = 0.010$). It was also marginally non-significant when this analysis was run on mean within-paper effect-size on all studies on the barn swallow, but only when sample was estimated as the minimum sample size within papers (Kendall's $\tau = -0.367$, $P = 0.052$). In contrast, it was non-significant in all the other cases, including the analyses run on all effect sizes separately (Kendall's $|\tau| \leq 0.362$, $P \geq 0.079$). Rosenthal's fail-safe number was always larger than the threshold of $5n + 10$ (where n is the original number of effect sizes included in each analysis) suggested by Rosenthal (1979) as a conservative value for a fail-safe calculation (Table S2). In addition, Rosenthal's fail-safe number was also *significantly* larger than the $5n + 10$ threshold in all cases except for indirect effects calculated on mean within-paper effect sizes (see Fragkos et al. 2014 for confidence intervals for Rosenthal's fail-safe number). Finally, the number of 'missing' studies computed using the 'trim and fill' method was larger than zero (but always ≤ 4) only in three cases, when the analyses were run on mean within-paper effect sizes (Table S2). However, in all these cases, the mean effect sizes recomputed by accounting for this potential source of bias were still significantly larger than zero (Table S2), so confirming that possible asymmetries in the effect size distributions should not have affected our results. We thus concluded that, although there was some suggestion of possible bias in a small number of analyses, overall publication bias should not have affected our conclusions on the barn swallow.

Discussion

This meta-analysis investigated the effects of livestock farming on the barn swallow and on other bird species typical of rural habitats. We found evidence of a positive effect of livestock farming on barn swallows, both when we considered the benefits provided by presence of livestock (direct effects) and those provided by features of the settings related to animal husbandry, like presence of rural buildings devoted to rearing animals or large extent of hayfields around farming sites (indirect effects). Direct benefits provided by livestock farming to barn swallows are probably related to the larger abundance and availability of insects and other invertebrates around settings where animals are reared (Møller 2001). However, barn swallow also seems to benefit from indirect effects of

livestock presence, even if this is to a lesser extent. Indeed, use of natural fertilizers derived from manure is positively associated with richness and abundance of arthropod taxa (McMahon et al. 2010). Moreover, arthropod-rich environments, like hayfields, meadows and grasslands in general, are usually larger around farms with livestock, offering foraging habitat for insectivorous birds (Ambrosini et al. 2002a; Evans et al. 2007; Grübler et al. 2010).

Livestock farming seems to affect barn swallow distribution, while we found no conclusive evidence of any effect of livestock farming on the reproduction of this species. The barn swallow is well-known to be strictly associated with livestock farming during the breeding period (Ambrosini et al. 2002a) and studies showed that colonies dramatically declined or went extinct after rearing or dairy activity ceased on a farm (Møller 2001; Ambrosini et al. 2002b, 2012; Willi et al. 2011). However, this meta-analysis highlighted that also presence of features related to animal husbandry at a farm may provide benefits to this species, even when livestock is not present (indirect effects). Interestingly, despite direct effects tended to have a larger influence on barn swallows than indirect ones, the difference between them was marginally non-significant. For example, presence of traditional cattle-sheds at a farm was found to enhance colony size of barn swallow, independently of the actual presence of livestock in them (Ambrosini et al. 2002a). In addition, a recent study observed that colony size increased with increasing extent of hayfields around the nesting site in farms without livestock, but not in those with livestock (Sicurella et al. 2014).

While it is known that disappearance of livestock farming may determine decline and even extinction of barn swallow colonies (Møller 2001; Ambrosini et al. 2012), our results suggest that this species may benefit also by the presence of other features typical of animal farming settings. Some of these features may persist in agricultural settings for years after livestock farming ceased. For example, cattle-sheds and stables often remain accessible to birds for nesting even after animal husbandry activities stopped on a farm. These results therefore suggest that, despite maintenance of livestock should be prioritized as a strategy for contrasting the decline observed in some population of these species (Ambrosini et al. 2012; Sicurella et al. 2014), maintenance of large hayfields and of rural buildings used by swallows for nesting may represent alternative strategies whenever maintenance of livestock farming is impractical due e.g. to economic reasons (Sicurella et al. 2014). In addition, persistence of indirect benefits may explain, at least partly, the observed latency in the response of barn swallow colonies to changes in the conditions of breeding habitats (Ambrosini et al. 2002b).

In contrast, our meta-analysis suggested that the effect of livestock on barn swallow reproduction was almost null, accounting for less than 0.1 % of the variance only. Such a small effect size on barn swallow reproduction was unexpected, particularly because this species nests almost exclusively in buildings (Turner 2006), particularly in those devoted to rearing animals. In addition, several studies indicated that barn swallows breed earlier, raise larger broods, produce nestlings of better quality, and have a higher probability of laying a second clutch in presence of livestock (Møller 2001; Ambrosini et al. 2002a). However, the size of the effects reported in these studies was small, probably because variation in clutch size and fledgling success, which are the components of breeding success recorded more often, is rather small in the barn swallow (the coefficient of variation of clutch size is 0.20 as assessed on a sample of 16,135 first clutches observed in Northern Italy in 1999–2014, while that of fledgling success is 0.55 as assessed on 18,941 first broods in the same area and period; R. Ambrosini unpublished data). Small variation in breeding success, in turn, may limit the options of observing large effects of livestock

farming on them. In addition, breeding success of barn swallows may be largely dependent on other effects unrelated to livestock farming, like presence of predators (e.g. mice *Mus musculus* and rats *Rattus* spp., Eurasian hobby *Falco subbuteo*, sparrowhawk *Accipiter nisus*) (Møller and Nielsen 1997). For instance, in Denmark the breeding pairs of barn swallows ‘just’ disappeared in the presence of barn owls *Tyto alba* that can lead to the extinction of large colonies in a year (A.P. Møller unpublished data).

We also investigated the hypothesis that the positive influence of livestock could be more evident at northern latitudes, where severe climatic conditions can determine larger variability in temperature and insect food availability during the breeding season (Møller 2001; Grübler et al. 2010). However, we did not find any significant relationship between effect sizes and latitude of study areas, either on the barn swallow or on the other species.

The size of the effects of livestock on other species of rural habitats was not significantly larger than zero in any case. This lack of effects of livestock on other bird species may be the result of the large heterogeneity of species considered. Indeed, the other bird species we considered included also not strictly insectivorous species, like the white stork, which were not expected to considerably benefit from the increasing insects abundance at setting with livestock. In addition, the barn swallow probably depends more strictly on human activities and buildings than other birds. Indeed, European populations of this species breed almost exclusively in buildings (Turner 2006). Unfortunately, paucity of studies did not allow investigations within ecological groups of species like e.g. insectivores or granivores.

Our meta-analysis therefore suggests that barn swallow populations seem to benefit from the presence of livestock. Such benefits seem to arise both from presence of domestic animals per se, and from availability of nesting sites and foraging areas. Reduction of animal husbandry may therefore have concurred to the recent decrease of populations of barn swallows (Møller 2001; Ambrosini et al. 2002a; Evans and Robinson 2004). In contrast, other bird species typical of rural habitats seem not to benefit from livestock farming. However, a positive effect of livestock on other species may have been masked by the large heterogeneity in the studies and in the species included in the analysis, as well as by the small number of papers that investigated this topic on other species.

During the last decades of the 20th centuries, the European Union introduced economic subsidies and agro-environmental schemes with the aim to improve agro-ecosystem biodiversity and to counteract the negative effect of agricultural intensification (Donald et al. 2001; Kleijn and Sutherland 2003). Farmland bird populations were expected to indirectly benefit from agro-environmental schemes, but precise evaluation of such effects has proven difficult, because effects seem to largely differ among species (Kleijn and Sutherland 2003; Hiron et al. 2013). Maintenance of livestock farming should therefore be included in conservation plans aiming at preserving barn swallow populations in rural habitats (Ambrosini et al. 2012; Sicurella et al. 2014).

Importantly, in contrast to the large amount of information available on the effect of arable field management on birds (see Fuller et al. 1995; Anderson et al. 2001; Vickery et al. 2001; Donald et al. 2002; Gregory et al. 2005; and references therein), our extensive search of the literature revealed that only 26 published papers focused on the effect of livestock farming practices on single bird species of rural habitats. Future studies on this topic are therefore welcome, also because all the species we considered except for the house martin are farmland birds, which are declining at a continental scale.

Acknowledgments We are grateful to Prof. Wim Van den Noortgate for advices on meta-analytic mixed models. Grzegorz Orłowski kindly provided additional information on sign of effect sizes not reported in

original papers. Comments by Dan Chamberlain and an anonymous reviewer greatly improved the quality of the paper. This research was funded by grant 2014-ATE-0437 to RA.

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